

PROPOSAL SUMMARY

PREPARED FOR

Mid-States Companies
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PROJECT

Eight (8) XM-4460 hammer mills for corn grinding — dry-grind ethanol plant
48–50 TPH per mill at 7,200 sq in screen area
Mills set on concrete; plenums hung below on the next floor
Air system (fans, filters, ducting) by W.D. Patterson — Mike King

REFERENCE

MCE-Q-2609-MIDSTATES-R1
Basis: Rough Rider hammer mill review of Sep 15, 2026 (C. Schaudt / J. Bliss / M. King) and MCE engineering of Sep 15, 2026
Validity: **through December 31, 2026** ·
Currency: USD
FOB MCE plant, Newkirk, OK

Midwest Custom Engineering is pleased to quote eight (8) XM-4460 extra-heavy-duty hammer mills with feeders, self-cleaning magnet adapters, maintenance gates, 350 FPM plenum chambers and plenum transfer screws for corn grinding at the Mid-States ethanol project. Each mill is rated 48–50 TPH on corn with 7,200 sq in of screen area and a 500 HP drive. TECO MAX-E1 500 HP main drive motors are priced as item 7, with Toshiba Global SD motors offered as an alternate; Mid-States may instead purchase the motors directly and ship them to MCE for mounting. The air-relief system is by W.D. Patterson and is not included here.

1. DESIGN BASIS

per Jason Bliss, MCE, Sep 15, 2026

PARAMETER	DESIGN VALUE	NOTES
Product	Whole corn — dry-grind ethanol	Grinding for fermentation; screen selection per Mid-States process
Number of mills	8 × XM-4460	Extra-heavy-duty, direct-drive, dual grinding chamber, gravity discharge
Capacity	48–50 TPH per mill	Based on 7,200 sq in screen area per mill
Screen size	#5 (5/64")	Two (4 total) sets of screens per mill included
Mill drive	500 HP, 1800 RPM, 460 V/3/60, TEFC premium efficiency, 5009/10/11 frame	TECO MAX-E1 as item 7; Toshiba Global SD offered as Option A; or supplied by Mid-States and shipped to MCE for mounting
Air relief through mill	9,360 CFM @ 6" SP per mill	Fans, dust filters, ducting and airlocks by W.D. Patterson
Feeder	14-row, 10" dia. nylon cup	20.72 RPM at 90% cup fill; auto self-cleaning magnet adapter
Maintenance gate	One per mill, electric-driven, between mill and plenum	1 HP Sumitomo right-angle gearmotor drive; two microswitches indicate gate open / gate closed
Plenum chamber	78" × 78" × 18'-2" long over flanges; 350 FPM design	MCE drawing XM4460-350PL Rev A; 3/16" A36 plate, 2" flat bar wraps every 24"; 35° hopper bottom to the transfer screw; hangs below mill floor — no support structure
Discharge	18" × 21'-0" transfer screw under each plenum	18'-0" flared inlet under the plenum hopper; single span, no hanger bearing; sized for 50 TPH ground corn; airlock and downstream conveyance by others
Installation	Mills set on concrete; plenums on floor below	Installation by others

2. SCOPE OF SUPPLY AND PRICING

USD · taxes and freight excluded

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
1	XM-4460 Hammer Mill — Extra Heavy Duty Mill Design Direct drive, dual grinding chamber, gravity discharge hammer mill: • Motor base extension; reversible top feed inlet; screen carriage • Screen size: #5 (5/64") — two (4 total) sets of screens included • Construction: mild steel with AR liners • 7,200 sq in screen area • Hammers: hard faced, heat treated — one (1) set included • HP range 300–600 HP; operating horsepower 500 HP • Capacity: 48–50 TPH corn; air relief 9,360 CFM @ 6" SP through the mill • Included: top feed inlet, mill/motor support base, three NEMA 4/12 limit switches, OSHA guard, vibration switch and rubber vibration pads • Motor: 500 HP, 1800 RPM, 460 V/3/60, TEFC premium efficiency, 1.15 SF — not included in the mill price; see item 7 (TECO) or Option A (Toshiba). Customer-supplied motors are shipped to MCE for mounting and alignment before the mill ships.	8	\$123,791.80	\$990,334.40
2	XM-10" AC 14-Row Nylon Cup Feeder with built-in Auto Self-Clean Magnet Adapter • 10" dia. nylon cup rotor, 14 rows; shaft extension, motor base and roller chain drive with safety shield • 1.608 cu ft/rev; 20.72 RPM at 90% cup fill • Combo unit built with auto self-clean magnet adapter: stainless steel front, stainless steel hinged magnet plate, air operated with two air cylinders and trash pans; SF300–50" plate magnet • Micro switch to indicate magnet door open • Gearhead motor quoted as item 3	8	\$41,631.70	\$333,053.60
3	Feeder Gearhead Motor 2 HP, 3 PH, 60 Hz, 460 V, TEFC gearhead motor, 40.7 RPM final output — sized for 20.72 RPM feeder speed through the roller chain drive.	8	\$4,059.00	\$32,472.00
4	Maintenance Gate — Electric Drive • Mounted between the hammer mill discharge and the plenum chamber • Push rods with rack and pinion gear seal off the hammer mill for maintenance and prevent dropped parts from entering the plenum • Electric drive in lieu of hand crank: 1 HP Sumitomo right-angle gearmotor, 3 PH, TEFC • Two micro switches indicate gate open and gate closed at the control panel • Adder for the electric gearmotor drive and microswitches over MCE's standard hand-crank gate — pending engineering pricing; base gate price shown is the hand-crank list price	8	\$10,835.00	\$86,680.00
5	Custom Plenum Chamber — 350 FPM, for XM-4460 (MCE dwg. XM4460-350PL Rev A) • 78" × 78" cross-section, 218-1/8" (18'-2") long over end flanges, 72" deep including 35° hopper bottom to a 19-13/16" wide discharge for the transfer screw trough • Constructed of 3/16" A36 plate, wrapped every 24" with 2" flat bar	8	\$44,567.60	\$356,540.80

ITEM	DESCRIPTION	QTY	UNIT PRICE	EXTENDED
	10" top flange with bolted 78" x 78" mill mounting opening at the mill end; 55" x 60-3/4" flanged opening at the opposite end for the dust filter / air-relief connection (filter by others) - Plenum hangs below the mill floor (mill sets on concrete) — no support structure, hangers or floor steel included - Approximate weight 2,500 lb each; exterior one shop coat grey enamel - Final length confirmed on approval drawings to suit mill placement			
6	Transfer Screw Conveyor — 18" dia. x 21'-0", for XM-4460 plenum 18" diameter screw conveyor, 21'-0" long inside trough ends, single span with no intermediate hanger bearing, to convey 50 TPH of ground corn at 35 PCF, control fed at 45% trough loading, horizontal. Price basis: SCC quote H51445AW1 — update on receipt of SCC quotation for this configuration. 18" RH sectional screw, 1/4" flights, full pitch, on 6" Sch 40 pipe, 3-7/16" shafts; Tem-U-Lac coupling bolts 18'-0" flared inlet trough, 3/16" plate, open to receive the full length of the plenum hopper discharge, with angle flange around the opening to bolt to the plenum - Shroud at the end of the 18'-0" opening, then 3'-0" of flanged U-trough with 12 GA gasketed covers to the discharge - Anti-friction tail trough end with WPS and lip seal; standard drive trough end with feet; flush end discharge spout; support feet; gaskets and bolt kits - Drive: 10 HP, 1750 RPM, 3/60/460 V TEFC motor; Dodge TA4207H25 taper-bushed reducer, class 2 gearing, 46 RPM output; drive shaft, motor mount, sheaves, belts and guard - Exterior: one shop coat grey enamel; general arrangement drawings for approval before fabrication	8	\$32,862.50	\$262,900.00
7	Main Drive Motor — TECO MAX-E1 500 HP TECO MAX-E1 premium efficiency, 500 HP, 1800 RPM, 460 V/3/60, TEFC, 5009/10/11 frame, cast iron, 1.15 SF. Mounted, aligned and guarded on the mill at MCE. Priced net — the project discount does not apply to motors. TECO quoted from stock (Round Rock, TX) — confirm availability of eight and lead time. Mid-States may instead purchase the motors directly and ship them to MCE for mounting at no additional charge.	8	\$31,336.00	\$250,688.00

Subtotal — items 1 to 6	\$2,061,980.80
Less 10% discount project discount, 8-mill order	(\$206,198.08)
Item 7 — TECO main drive motors (8), priced net	\$250,688.00
Total — eight (8) mill lines as quoted	\$2,106,470.72

Unit prices are quoted for a single order of eight (8) lines. The project discount applies to items 1–6; motors (item 7 and Option A) are priced net.

OPTIONS AND ADDERS

OPTION	DESCRIPTION	QTY	UNIT PRICE	NET ADDER
A	Toshiba Global SD 500 HP main drive motors, in lieu of item 7 Toshiba Global SD premium efficiency, 500 HP, 1800 RPM, 460 V/3/60, TEFC, 5009/10/11 frame, 1.15 SF (Toshiba part 5004FTAB41E-A). Mounted, aligned and guarded on the mill at MCE. Factory lead time approximately 13 weeks (December 20, 2026 on a September order). Adder shown is the net difference over item 7 for eight motors.	8	\$47,118.00	\$126,256.00 \$15,782.00 each
B	Plenum chambers fabricated by Mid-States Mid-States may elect to fabricate the plenum chambers to MCE drawings. MCE furnishes fabrication drawings and the mill mounting flange details; items 1-4, 6 and 7 are unchanged. Confirm whether MCE supplies the mill mounting flange/adaptor with this option.			Deduct item 5

3. FURNISHED BY OTHERS

- Air-relief system for the mills — fans, dust control filters, ducting, airlocks and explosion protection — by W.D. Patterson (Mike King), sized to 9,360 CFM @ 6" SP per mill as stated above
- Mill drive motors if supplied by Mid-States (see item 7); motor starters, VFDs and motor control
- Controls, PLC programming and instrumentation beyond the limit switches, vibration switches and magnet micro switches listed
- Concrete, foundations, floor openings, anchor bolts and any support steel for the plenums
- Infeed to the feeders, and conveyance from the transfer screw discharge
- Compressed air to the magnet adapters (air cylinders)
- Installation, millwright, electrical and start-up labor

4. FREIGHT, DELIVERY AND SCHEDULE

ITEM	BASIS
Freight	FOB MCE plant, Newkirk, OK; freight to Mid-States site quoted at time of shipment or shipped freight collect.
Delivery	__ weeks after receipt of order, approval drawings and customer-supplied motors (if applicable). Order placement planned by Mid-States for December 2026; MCE will reserve fabrication capacity on receipt of a signed proposal.
Approval drawings	Plenum chamber per MCE drawing XM4460-350PL Rev A (Sep 17, 2026); general arrangement drawings for the plenums, maintenance gates and transfer screws issued for Mid-States approval before fabrication, with plenum length confirmed to suit mill placement.
Installation	By others. MCE start-up assistance and operator training available at MCE's standard daily rate plus expenses.

5. COMMERCIAL TERMS

Payment: 50% down with order, balance before shipment. Prices are in US dollars and exclude sales tax, freight and installation. Pricing is valid **through December 31, 2026**; steel and motor pricing are subject to adjustment for orders placed after that date. Warranty per MCE standard terms and conditions. This proposal covers the equipment listed in Section 2 only; the air system is quoted separately by W.D. Patterson.

PREPARED BY

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FOR MID-STATES COMPANIES

Accepted: _____

Name / Title: _____

Date: _____ PO #: _____